

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 2 – Agile Sprint Planning & User Story Workshop
Weightage:	20%	Total Marks:	25
CO1 Statement:	<i>Apply advanced methodologies and agile project management techniques to manage software projects effectively</i>		
Student Name:		Reg. No.:	
Date:		Duration:	60 minutes

Code of Conduct

I E Ajay Velmail (192311385) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____ E Ajay Velmail _____

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

- 1. Identify the Project Vision and Stakeholders**
 - Define the project's vision, objectives, and identify all key stakeholders involved.
- 2. Create Epics and User Stories**
 - Break down the requirements into Epics and formulate well-structured User Stories using the format:
As a <user>, I want <feature>, so that <benefit>.
- 3. Define Acceptance Criteria**
 - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
- 4. Prioritize the Product Backlog**
 - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
- 5. Plan the Sprint Backlog**
 - Select high-priority user stories for the first sprint and divide them into actionable development tasks.
- 6. Estimate Effort Using Story Points**
 - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.
- 7. Present the Sprint Goal and Expected Deliverables**
 - Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification	/5
Epics & User Stories	/5
Acceptance Criteria	/5
Product Backlog Prioritization & Sprint Planning	/5
Story Point Estimation, Sprint Goal & Presentation	/5
TOTAL	/25

AGILE SPRINT PLANNING & USER STORY WORKSHOP

Project: Cold-Chain Logistics Integrity Platform

1. PROJECT VISION AND STAKEHOLDER IDENTIFICATION

1.1 Project Vision

The vision of the **Cold-Chain Logistics Integrity Platform** is to develop a scalable, secure, and intelligent real-time monitoring system that protects temperature-sensitive food and pharmaceutical products throughout transportation and storage.

The platform will use IoT sensors to continuously collect **temperature, humidity, GPS location, door status, and refrigeration-unit health data**. The collected data will be processed in real time to detect environmental excursions, identify potential shipment failures, notify stakeholders, and automatically generate compliance documentation.

The system aims to transform traditional cold-chain monitoring from a **reactive, post-delivery process into a proactive, real-time and predictive system**.

1.2 Project Objectives

The project aims to:

1. Continuously collect IoT sensor data.
2. Monitor shipment conditions in real time.
3. Detect environmental excursions automatically.
4. Apply product-specific environmental thresholds.
5. Generate immediate alerts for critical conditions.
6. Predict possible refrigeration or shipment failures.
7. Track shipment location using GPS.
8. Provide real-time operational dashboards.
9. Maintain historical telemetry and audit trails.
10. Generate automated compliance reports.
11. Support thousands of connected vehicles and devices.
12. Provide secure, reliable, and highly available monitoring.

1.3 Key Stakeholders

Stakeholder	Role
Logistics Manager	Monitors fleet, shipments, incidents, and operational performance
Operations Manager	Responds to alerts and coordinates corrective actions
Driver	Transports shipments and responds to relevant alerts
Warehouse Operator	Monitors storage conditions and loading/unloading activities
Food Manufacturer/Distributor	Ensures food products remain within required conditions
Pharmaceutical Manufacturer/Distributor	Ensures pharmaceutical products maintain required conditions

Quality/Compliance Officer	Reviews compliance data and generates documentation
IoT Administrator	Registers, configures, and monitors sensors
System Administrator	Manages users, organizations, access, and system configuration
Data/ML Engineer	Develops predictive failure and anomaly-detection models
Regulatory Authority	Verifies compliance documentation when required
Customer/Receiver	Receives products and benefits from verified shipment integrity

2. EPICS AND USER STORIES

The major requirements are divided into **8 Epics**.

Epic 1 – User and Organization Management

US-01: User Login

As a system user,

I want to securely log into the platform,

so that I can access the features authorized for my role.

US-02: Role-Based Access

As an administrator,

I want to assign roles and permissions to users,

so that users can access only the information and functions relevant to them.

Epic 2 – IoT Sensor Management

US-03: Register Sensor

As an IoT administrator,

I want to register a new IoT sensor in the platform,

so that the sensor can send data to the monitoring system.

US-04: Monitor Sensor Health

As an IoT administrator,

I want to view the health and connectivity status of sensors,
so that faulty or disconnected devices can be identified quickly.

Epic 3 – Real-Time Telemetry Monitoring

US-05: Receive Sensor Data

As a logistics operator,

I want the platform to receive real-time temperature and environmental data,
so that I can continuously monitor shipment conditions.

US-06: View Live Shipment Conditions

As a logistics manager,

I want to view the current temperature, humidity, GPS location, door status, and refrigeration health of shipments,

so that I can understand the current condition of the logistics network.

Epic 4 – Threshold and Excursion Management

US-07: Configure Product Thresholds

As a quality officer,

I want to configure acceptable environmental ranges for each product,
so that the system can identify conditions that may compromise product quality.

US-08: Detect Temperature Excursions

As a quality officer,

I want the system to automatically detect temperature excursions,
so that corrective action can be initiated before products are compromised.

US-09: Detect Door and Equipment Events

As an operations manager,

I want to receive events for abnormal door openings and refrigeration-unit conditions,
so that I can investigate possible causes of environmental excursions.

Epic 5 – Alert and Notification Management

US-10: Receive Critical Alerts

As a logistics manager,

I want to receive immediate alerts when critical environmental conditions occur,
so that I can take corrective action before the shipment becomes unusable.

US-11: Alert Escalation

As an operations manager,

I want unresolved critical alerts to be escalated to responsible personnel,
so that important incidents are not ignored.

Epic 6 – Shipment and GPS Tracking

US-12: Track Shipment Location

As a logistics manager,

I want to view the real-time GPS location of a shipment,
so that I can monitor its movement and identify delays.

US-13: Associate Telemetry with Shipment

As a logistics operator,

I want sensor data to be associated with a specific shipment and vehicle,
so that environmental events can be traced to the correct shipment.

Epic 7 – Predictive Analytics

US-14: Predict Refrigeration Failure

As an operations manager,

I want the system to identify patterns indicating potential refrigeration failure,
so that preventive action can be taken before the shipment is compromised.

US-15: Identify High-Risk Shipments

As a logistics manager,

I want the system to identify shipments with elevated environmental risk,
so that operational resources can be prioritized effectively.

Epic 8 – Analytics, Compliance and Audit

US-16: View Historical Data

As a quality officer,

I want to view historical shipment conditions,
so that I can investigate previous incidents and analyze performance.

US-17: Generate Compliance Report

As a compliance officer,

I want to automatically generate shipment compliance reports,
so that regulatory documentation can be prepared efficiently.

US-18: Maintain Audit Trail

As a compliance officer,

I want all important events and user actions to be recorded,
so that the system provides traceable evidence for audits.

3. ACCEPTANCE CRITERIA

The assessment requires acceptance criteria to be **specific, measurable, testable, and aligned with the user stories**, preferably using the Given–When–Then format.

The following acceptance criteria are defined for the high-priority stories selected for Sprint 1.

US-01 – User Login

Acceptance Criteria

Given a registered user has valid credentials

When the user enters the correct username and password

Then the system shall authenticate the user and display the appropriate dashboard.

Given a user enters incorrect credentials

When the login request is submitted

Then the system shall reject the login and display an appropriate error message.

US-03 – Register Sensor

Acceptance Criteria

Given an administrator is authenticated

When valid sensor information is submitted

Then the system shall register the sensor successfully.

Given a sensor ID already exists

When the administrator attempts to register it again

Then the system shall reject the duplicate registration.

US-05 – Receive Sensor Data

Acceptance Criteria

Given a registered sensor has network connectivity

When the sensor transmits valid telemetry

Then the platform shall receive and store the telemetry with a timestamp.

Given invalid telemetry is received

When the platform validates the data

Then the invalid record shall be rejected or flagged for investigation.

US-06 – View Live Shipment Conditions

Acceptance Criteria

Given a shipment has active sensors

When the user opens the monitoring dashboard

Then the latest available temperature, humidity, location, door status, and refrigeration status shall be displayed.

Given new telemetry is received

When the dashboard refreshes

Then the latest valid values shall be displayed.

US-07 – Configure Product Thresholds

Acceptance Criteria

Given a quality officer is authorized to configure thresholds

When valid minimum and maximum environmental values are entered

Then the system shall save the threshold configuration for the selected product.

Given an invalid threshold range is entered

When the configuration is submitted

Then the system shall reject the configuration and display an error.

US-08 – Detect Temperature Excursions

Acceptance Criteria

Given a shipment has a configured temperature range

When incoming temperature data exceeds the configured limit

Then the platform shall create a temperature-excursion event.

Given a critical excursion is detected

When the event is created

Then the relevant stakeholder shall receive an alert.

US-10 – Receive Critical Alerts

Acceptance Criteria

Given a critical environmental event occurs

When the event is detected

Then the platform shall generate a high-priority alert.

Given a critical alert has been generated

When the responsible user receives the notification

Then the user shall be able to view the shipment, condition, timestamp, and location associated with the incident.

US-12 – Track Shipment Location

Acceptance Criteria

Given a shipment has an active GPS device

When valid GPS data is received

Then the platform shall update the shipment's current location.

Given a shipment is displayed on the dashboard

When the user selects it

Then the latest known location shall be displayed.

US-17 – Generate Compliance Report

Acceptance Criteria

Given a completed shipment has recorded telemetry

When an authorized compliance officer requests a report

Then the platform shall generate a report containing relevant environmental history and excursion events.

Given the shipment has no recorded excursions

When the report is generated

Then the report shall indicate that no environmental excursions were recorded.

4. PRODUCT BACKLOG PRIORITIZATION

The product backlog is prioritized according to:

- Business value
- Customer and regulatory importance
- Risk reduction
- Technical dependencies
- Operational importance
- Implementation feasibility

The assessment specifically requires logical prioritization based on business value, customer needs, and project dependencies.

Priority Levels

P1 – Must Have: Essential for the basic working system.

P2 – Should Have: Important but can follow the core functionality.

P3 – Could Have: Valuable enhancement that is not essential for the first release.

Product Backlog

Priority	ID	User Story	Epic	Story Points
P1	US-01	Secure user login	User Management	3
P1	US-03	Register IoT sensor	Sensor Management	5
P1	US-05	Receive real-time telemetry	Telemetry	8
P1	US-07	Configure product thresholds	Threshold Management	5
P1	US-08	Detect temperature excursions	Excursion Management	8
P1	US-10	Generate critical alerts	Notification	5
P1	US-06	View live shipment conditions	Monitoring	8
P1	US-13	Associate telemetry with shipment	Shipment Management	5
P2	US-12	Track shipment GPS location	GPS Tracking	5
P2	US-04	Monitor sensor health	Sensor Management	5
P2	US-09	Detect door/equipment events	Event Management	5
P2	US-17	Generate compliance reports	Compliance	8
P2	US-16	View historical data	Analytics	5
P2	US-18	Maintain audit trail	Compliance	5
P2	US-11	Alert escalation	Notification	5
P3	US-14	Predict refrigeration failure	Predictive Analytics	13
P3	US-15	Identify high-risk shipments	Predictive Analytics	8
P3	US-02	Advanced role-based access	User Management	5

5. SPRINT 1 PLANNING

5.1 Sprint Duration

Sprint Duration: 2 weeks

A two-week sprint is selected because it provides enough time to develop and test core functionality while maintaining frequent feedback.

5.2 Sprint Objective

Sprint 1 will establish the **technical foundation of the Cold-Chain Logistics Integrity Platform** and deliver the minimum functionality required to register sensors, receive telemetry, configure environmental thresholds, detect basic temperature excursions, and notify users.

5.3 Sprint Backlog

The following high-priority stories are selected for Sprint 1.

ID	User Story	Story Points
US-01	Secure user login	3
US-03	Register IoT sensor	5
US-05	Receive real-time telemetry	8
US-07	Configure product thresholds	5
US-08	Detect temperature excursions	8
US-10	Generate critical alerts	5

Total = 34 Story Points

The sprint focuses on core dependencies rather than attempting to implement advanced predictive analytics immediately.

5.4 Sprint Task Breakdown

US-01 – Secure User Login

Tasks:

1. Design login interface.
2. Create user authentication API.
3. Implement credential validation.
4. Implement session/token handling.
5. Add authentication error handling.
6. Perform authentication testing.

US-03 – Register IoT Sensor

Tasks:

1. Design sensor registration form.

2. Create sensor database structure.
3. Implement sensor registration API.
4. Validate sensor IDs.
5. Prevent duplicate sensor registration.
6. Test sensor registration.

US-05 – Receive Real-Time Telemetry

Tasks:

1. Define telemetry data structure.
2. Develop telemetry ingestion API/service.
3. Validate incoming sensor data.
4. Add timestamps to telemetry records.
5. Store telemetry in the database.
6. Test continuous telemetry ingestion.
7. Test invalid-data handling.

US-07 – Configure Product Thresholds

Tasks:

1. Design threshold configuration interface.
2. Create product-threshold data model.
3. Implement threshold configuration API.
4. Validate minimum and maximum values.
5. Associate thresholds with products.
6. Test threshold configuration.

US-08 – Detect Temperature Excursions

Tasks:

1. Develop threshold evaluation logic.
2. Compare incoming telemetry against product thresholds.
3. Create excursion-event records.
4. Assign excursion severity.
5. Store excursion timestamps and values.
6. Test normal and abnormal temperature scenarios.

US-10 – Generate Critical Alerts

Tasks:

1. Design alert data model.
2. Create alert-generation service.
3. Connect excursion events to alerts.
4. Assign alert priority.
5. Implement notification mechanism.
6. Display alert information to authorized users.
7. Test critical alert generation.

6. STORY POINT ESTIMATION

6.1 Estimation Technique

The team uses **Planning Poker with the Fibonacci sequence**:

1, 2, 3, 5, 8, 13, 21...

Story points represent relative effort and complexity rather than exact hours.

The assessment permits estimation using approaches such as Fibonacci or Planning Poker and requires justification for the estimates.

6.2 Sprint 1 Estimation

User Story	Points	Justification
US-01 – User Login	3	Relatively straightforward authentication functionality with limited integration
US-03 – Sensor Registration	5	Requires database, validation, API, and sensor-management logic
US-05 – Telemetry Ingestion	8	High complexity because it involves continuous data ingestion, validation, storage, and scalability considerations
US-07 – Product Thresholds	5	Requires configurable product rules, validation, and database integration
US-08 – Excursion Detection	8	Requires real-time rule evaluation and event generation
US-10 – Critical Alerts	5	Requires integration with the excursion engine and notification mechanism

Total Sprint Estimate

3 + 5 + 8 + 5 + 8 + 5 = 34 Story Points

The larger estimates are assigned to telemetry ingestion and excursion detection because they involve real-time processing and are technically more complex than basic CRUD or authentication functionality.

7. SPRINT GOAL AND EXPECTED DELIVERABLES

7.1 Sprint Goal

"Build the core monitoring foundation that securely registers IoT sensors, receives real-time telemetry, applies product-specific environmental thresholds, detects temperature excursions, and generates critical alerts."

The Sprint Goal provides a clear direction for the development team and ensures that the first sprint produces a meaningful increment of the cold-chain platform.

7.2 Expected Sprint Deliverables

At the end of Sprint 1, the team is expected to deliver:

1. Working Authentication

Users can securely access the platform.

2. IoT Sensor Registration

Administrators can register and manage basic sensor information.

3. Telemetry Ingestion

The platform can receive and store real-time sensor data.

4. Product Threshold Configuration

Authorized users can configure acceptable environmental limits.

5. Temperature Excursion Detection

The system automatically identifies temperature values outside configured limits.

6. Critical Alerts

Relevant users receive alerts when critical environmental excursions occur.

7. Basic Testing

Unit and integration tests verify the functionality of the completed components.

8. DEFINITION OF DONE

A user story will be considered **Done** only when:

1. The required functionality has been implemented.
2. Code has been reviewed.
3. Unit testing has been completed.
4. Integration testing has been completed where applicable.
5. Acceptance criteria have been satisfied.
6. Security checks have been performed for relevant functionality.
7. No critical defects remain.
8. The functionality is integrated into the sprint build.
9. Required technical documentation has been updated.
10. The Product Owner accepts the completed story.

9. SPRINT WORKFLOW

The Sprint will follow the following workflow:

Product Backlog

↓

Sprint Planning

↓

Select High-Priority User Stories

↓

Break Stories into Tasks



Development



Continuous Testing



Daily Scrum



Sprint Review



Product Owner Feedback



Sprint Retrospective



Next Sprint

10. DAILY SCRUM ACTIVITIES

During Sprint 1, the development team will conduct a short Daily Scrum.

Each team member answers:

1. What did I complete yesterday?
2. What will I work on today?
3. Are there any blockers?

Example

Developer: Completed sensor registration API. Today I will work on telemetry ingestion.

Tester: Completed authentication testing. Today I will prepare telemetry validation test cases.

IoT Engineer: Completed sensor communication prototype. Today I will integrate it with the telemetry service.

Scrum Master: No major blockers reported.

11. SPRINT REVIEW

At the end of Sprint 1, the team demonstrates the working increment to the Product Owner and stakeholders.

The demonstration should show:

1. User login.
2. Sensor registration.
3. Incoming telemetry.
4. Stored sensor data.
5. Product threshold configuration.
6. Temperature excursion detection.
7. Critical alert generation.

Stakeholder feedback will be recorded and converted into new or modified backlog items for subsequent sprints.

12. SPRINT RETROSPECTIVE

The team reviews the sprint using three categories:

What Went Well?

- Core telemetry pipeline was implemented.
- Sensor registration was completed.
- Temperature excursion logic was successfully tested.
- Stakeholders received an early working prototype.

What Could Be Improved?

- Improve telemetry validation.
- Reduce integration delays.
- Improve dashboard usability.
- Increase automated test coverage.

Action Items

- Introduce additional automated tests.
- Improve API documentation.
- Conduct earlier integration testing.
- Refine alert rules based on stakeholder feedback.

13. FUTURE SPRINT PLAN

After Sprint 1, future sprints can progressively expand the platform.

Sprint	Major Features
Sprint 1	Authentication, sensor registration, telemetry, thresholds, excursion detection, alerts
Sprint 2	Live monitoring dashboard, GPS tracking, door status, refrigeration health
Sprint 3	Historical analytics, audit trail, compliance reporting
Sprint 4	Predictive failure detection and risk scoring
Sprint 5	Advanced analytics, mobile support, performance and scalability improvements
Sprint 6	Security hardening, deployment, acceptance testing, and production readiness

CONCLUSION

The Agile Sprint Planning and User Story Workshop provides a structured approach for developing the Cold-Chain Logistics Integrity Platform incrementally.

The project vision focuses on providing real-time, reliable, and proactive monitoring of temperature-sensitive food and pharmaceutical shipments. The requirements are divided into meaningful **Epics and User Stories**, following the standard Agile format of:

"As a , I want , so that ."

Acceptance criteria are defined using the **Given–When–Then** approach, making each requirement measurable and testable.

The Product Backlog is prioritized according to business value, operational importance, regulatory requirements, technical dependencies, and risk. Sprint 1 focuses on the most fundamental capabilities: authentication, IoT sensor registration, telemetry ingestion, threshold configuration, excursion detection, and critical alerts.

Using **Planning Poker and Fibonacci-based story points**, Sprint 1 is estimated at **34 story points**. The sprint goal is to establish the core monitoring foundation and deliver a working increment that demonstrates real-time cold-chain monitoring.

The Agile approach allows the platform to evolve through continuous stakeholder feedback, iterative development, testing, and improvement. Future sprints can progressively introduce

GPS tracking, compliance reporting, analytics, predictive failure detection, and advanced AI capabilities.

Therefore, the proposed sprint plan provides a **realistic, measurable, stakeholder-focused, and technically feasible roadmap** for developing the Cold-Chain Logistics Integrity Platform using Agile software engineering practices.